

MYP 1 Design

Handout 5: Repeating Actions (Control Blocks)

Learning Objective

By the end of this lesson, you will be able to:

- Explain the purpose of Control blocks.
- Use **Wait**, **Repeat**, and **Forever** blocks.
- Create programs that repeat actions.
- Build simple animations using loops.

What are Control Blocks?

Control blocks help control the flow of a program.

They tell the computer **when to wait**, **how many times to repeat an action**, or **whether to keep doing something forever**.

Think of them like traffic signals --- they control what happens and when.

Meet the Control Blocks

Block	Purpose
Wait	Pauses the program for a short time
Repeat	Repeats an action a certain number of times
Forever	Repeats an action continuously until the program is stopped

1. Wait Block: The **Wait** block pauses the program before moving to the next instruction.

Activity

When Green Flag Clicked

↓

Say "Hello!"

↓

Wait 2 Seconds

↓

Say "Welcome to Coding!"

Challenge

- Change the waiting time to **1 second**.
- Change it to **5 seconds**.
- What do you notice?

2. Repeat Block: The **Repeat** block repeats the instructions inside it a specific number of times.

Activity

When Green Flag Clicked

↓

Repeat (5)

Move 20 Steps

Challenge

- Change Repeat to **10**.
- Change Move to **10 Steps**.
- What changes?

3. Forever Block: The **Forever** block keeps repeating the instructions until the program is stopped.

Activity

When Green Flag Clicked

↓

Forever

Turn 15 Degrees

Challenge

- Add **Move 5 Steps** inside the Forever block.
- What happens?

Activity 4: Create a Dancing Sprite

Use:

- When Green Flag Clicked
- Forever
- Move
- Turn
- Wait

Make your sprite dance continuously.

Activity 5: Flashing Message

Create this program:

When Green Flag Clicked

↓

Forever

Say "Hello!"

Wait 1 Second

Say "Welcome!"

Wait 1 Second

Observe

What happens to the message?

Activity 6: Walking Animation

Make the sprite:

- Move 20 steps.
- Wait 0.5 seconds.
- Repeat 10 times.

Can you make it look like the sprite is walking?

Debugging Challenge: A student wrote the below program

When Green Flag Clicked

↓

Move 100 Steps

↓

Forever

Turn 15 Degrees

The student wanted the sprite to **move and turn continuously**.

Question

What is the mistake?

How can you fix it?

Hint: Which block should be inside the **Forever** loop?

Mini Challenge

Create your own animation using:

- Events
- Motion
- Looks
- Control

Your program must:

- Start with the Green Flag.
- Move the sprite.
- Repeat actions.
- Include a Wait block.
- Display a message.

Quick Summary/Recap

Control Block	Purpose
Wait	Pauses the program
Repeat	Repeats actions a fixed number of times
Forever	Repeats actions until stopped

Reflection

1. Which Control block did you use the most?
2. What is the difference between **Repeat** and **Forever**?
3. Why is the **Wait** block useful?
4. Which activity did you enjoy the most?
5. What new thing did you learn today?

Homework

Create a simple animation using:

- **When Green Flag Clicked**
- **Repeat**
- **Forever**
- **Wait**
- **Motion** blocks

Your animation should run smoothly and repeat at least one action.